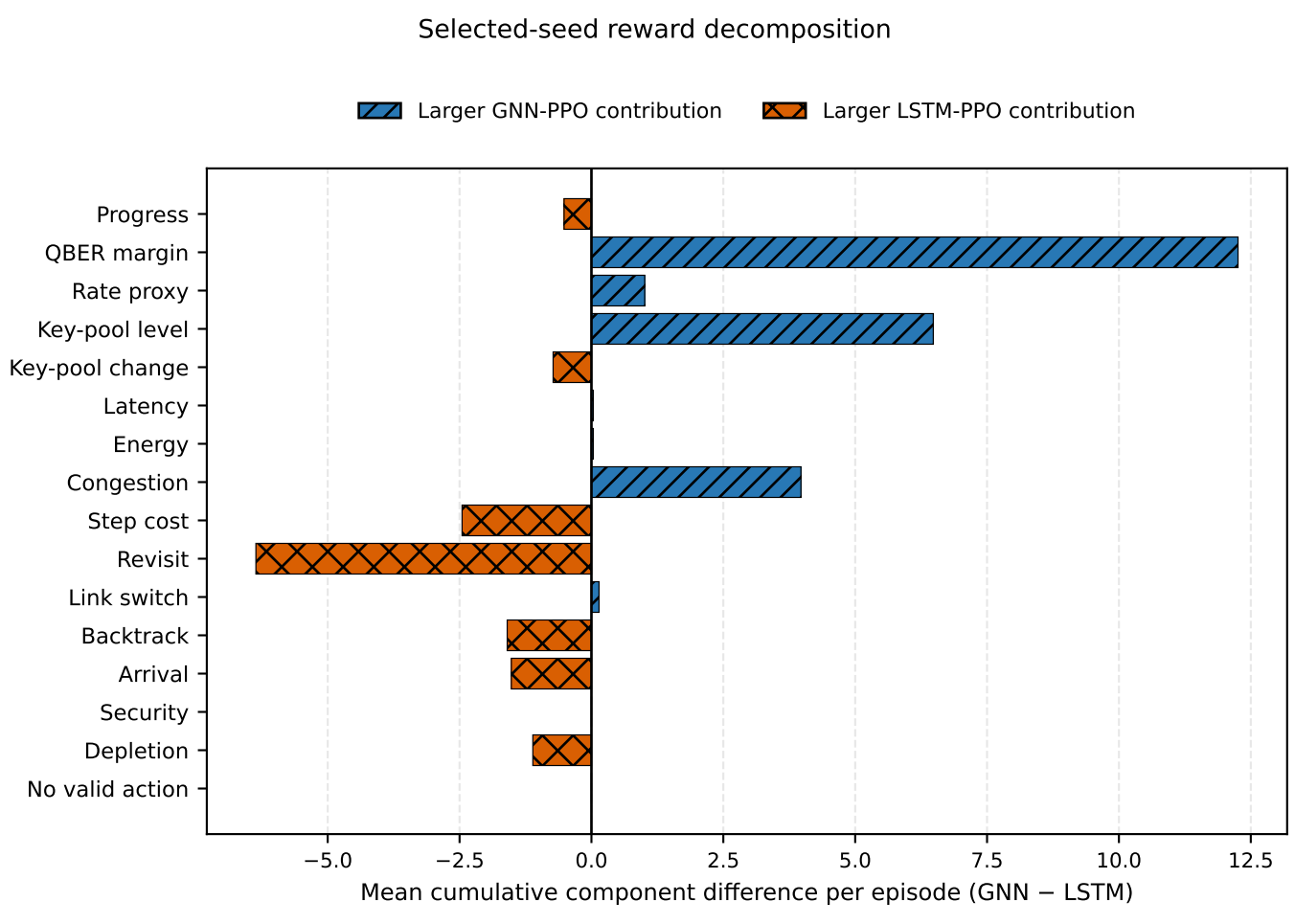
Selected-seed reward decomposition
Larger GNN-PPO contribution
Larger LSTM-PPO contribution
Progress
QBER margin
Rate proxy
Key-pool level
Key-pool change
Latency
Energy
Congestion
Step cost
Revisit
Link switch
Backtrack
Arrival
Security
Depletion
No valid action
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
Mean cumulative component difference per episode (GNN − LSTM)